Abdalla Samir Khalifa

Java Developer

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Disciplined Software Engineer committed to continuous learning and driven to contribute to business growth through high performance, maintainable systems. eager to scale my expertise to deliver the architectural integrity and operational efficiency required for our collective success.

Education

Bachelor's degree in Management Information Systems, *Higher Institute of Management Science* June 2025
GPA: 3.0/4.0 (Very Good).

Skills

Languages : Java, SQL, PL/SQL.
Frameworks: Spring Framework, Spring Boot, Spring Web, Spring Security, JWT, Jakarta Persistence (JPA), Hibernate, Spring Data JPA.
Database: MySQL, PostgreSQL.
Developer Tools: Docker, Git, Linux, Maven.
Concepts & Methodologies: Design Patterns, RESTful API Design, Database Design, Query Optimization, Normalization/De-Normalization, Performance Tuning (Indexing), System Analysis and Design .
Other: JavaScript, CSS, HTML.

Projects

Food Delivery Backend Platform

Tech Stack: Java 21, Spring Boot 3+, Spring Data JPA, PostgreSQL, Spring Security, Spring SMTP, Docker, Docker Compose, Maven, Git.

- **Co-developing** a feature-modularized platform (**Restaurants**, **Carts**, **Orders**) in a 3-person team using a strict **Main/Dev/Feature** Git branching model.
- Built the foundational relational schema, DDL scripts, and implemented core **Spring Data JPA** entities and repositories.
- **Containerized** the application and database infrastructure utilizing **Docker & Docker Compose** to establish a uniform development environment for the team.

E-Commerce Backend Platform

Tech Stack: Java 17, Spring Boot 3+, Spring Data JPA, Spring Security, Spring SMTP, Docker, Docker Compose, Maven, Git.

- Developed a **Spring Boot 3** e-commerce REST API with **Spring Security**, enabling product browsing, cart management, and secure checkout. Integrated **Spring SMTP** to automate welcome and order confirmation emails.
- Integrated **Spring SMTP** to automate welcome and order-confirmation emails on registration and checkout events.
- Secured API endpoints using **JWT** and **RBAC**, enforcing role-based access control to protect sensitive data.
- Containerized the application stack using **Docker Compose** for simplified deployment of API, and **PostgreSQL**.
- Built a global exception handler standardizing error responses across all Spring Boot API endpoints.

Database Design & Optimization

Tech Stack: PostgreSQL, MySQL, Git.

- Engineered a **PostgreSQL** relational schema for 50M+ records, used **EXPLAIN ANALYZE** to identify bottlenecks and applied B-Tree indexing, reducing query latency by **98.8%** (613ms → 7ms)
- Built a denormalization layer maintained by **PL/pgSQL** triggers, achieving a **99.9% performance improvement** on analytical revenue queries
- Validated schema decisions through MySQL benchmarking on 6M records, demonstrating a **119% increase in QPS** via composite indexing

Language

Arabic (fluent speaker) **English** (Very Good)